

Program : Diploma in Computer Application and Business Management	
Course Code : 2257	Course Title: Introduction to C Programming
Semester : 2	Credits: No Credit
Course Category: Foundation Course	
Periods per week: 4 (L:1, T:0, P:3)	Periods per semester: 60

Course Objectives:

- To provide a hands-on experience in programming concepts using C Programming Language and to develop programs using arrays, functions, structure and files.
- To enhance the problem solving skills of students.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic programming concepts		Introduction to Python	1

Course Outcomes:

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate sequential and control structures in C with simple problems	9	Applying
CO2	Use one and two dimensional arrays in C to solve real world problems.	18	Applying
CO3	Apply the concepts – functions, storage classes and recursion to solve problems.	15	Applying
CO4	Make use of Pointers and structures to solve real world problems effectively.	15	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate sequential and control structures in C with simple problems.		
M1.01	Develop simple C programs using decision control structures in C	3	Applying
M1.02	Implement C Programs using loop control structures.	6	Applying
CO2	Use one and two dimensional arrays in C to solve real world problems.		
M2.01	Develop C program to solve problems using one dimensional arrays – (like average of n nos, prime nos in a list, largest number, second largest etc)	3	Applying
M2.03	Implement linear search and bubble sort algorithms in C.	3	Applying
M2.04	Develop C programs to solve problems using two dimensional arrays – (like matrix operations)	6	Applying
M2.06	Implement C programs using strings	6	Applying
	Lab Exam – I	1½	
CO3	Apply the concepts – functions, storage classes and recursion to solve problems.		
M3.01	Implement C programs using user defined functions and recursion	6	Applying
M3.02	Implement Programs in C to illustrate passing of arrays to functions.	3	Applying
M3.03	Develop binary search and quick sort algorithms using recursion.	6	Applying

CO4	Make use of Pointers and structures to solve real world problems effectively.		
M4.01	Develop programs using pointers.	3	Applying
M4.02	Implement programs by passing arrays as pointers to functions	3	Applying
M4.03	Develop programs in C using structures	3	Applying
M4.04	Open Ended Experiments	6	Applying
	Lab Exam – II	1½	

Text / Reference:

T/R	Book Title/Author
R1	Balagurusamy E, Programming in ANSI C 7 th Edn, McGraw Hill.
R2	Yashavant Kanetkar, Let Us C , BPB Publications.
R3	Paul J. Deitel, Harvey Deitel, C How to Program , 8th edition, Pearson Education, 2015.
R4	Brian W. Kernighan, Dennis M. Ritchie, The C Programming Language , Prentice Hall of India, 2 nd Edition.
R5	Herbert Schild, C: The Complete Reference , McGraw Hill, 4th Edition, 2002.

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/106104128/
2	https://www.programiz.com/c-programming
3	https://www.tutorialspoint.com/cprogramming/index.htm